Week :10 (2-6 November 2026)

الأسبوع :#من - إلى

Subject/الموضوع: Advanced Physics **Unit Title/عنوان الوحدة: Motion in Two Dimensions - Torque and Rotational Equilibrium** **Grade/الصف: 12**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1 الحصة الأولى	Learning Target: I can explain torque as the turning effect of a force and identify the pivot, lever arm and rotational direction. (DOK 2)	2.1 Clarify vocabulary and symbols using visuals and supports. 3.2 Highlight patterns and key features using diagrams and worked examples.	Please check students' Teams for lesson materials and updates.
Lesson 2 الحصة الثانية	Learning Target: I can calculate torque using $\tau = rF \sin\theta$ and explain the effect of force, distance and angle. (DOK 3)	2.1 Clarify vocabulary and symbols using visuals and supports. 3.2 Highlight patterns and key features using diagrams and worked examples.	Please check students' Teams for lesson materials and updates.
Lesson 3 الحصة الثالثة	Learning Target: I can calculate net torque and determine whether a system is in rotational equilibrium, justifying my sign convention. (DOK 3)	1.1 Flexible display of information (size, colour, font, layout). 2.1 Clarify vocabulary and symbols using visuals and supports.	Please check students' Teams for lesson materials and updates.
Lesson 4 الحصة الرابعة	Learning Target: I can investigate and evaluate how force and lever-arm distance influence torque in a practical engineering context. (DOK 3)	1.1 Flexible display of information (size, colour, font, layout). 2.1 Clarify vocabulary and symbols using visuals and supports. 3.2 Highlight patterns and key features using diagrams and worked examples.	Please check students' Teams for lesson materials and updates.
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Bring a calculator, ruler, pencil and Physics notebook. Follow safety instructions during lever investigations.		

Week # 10 #الأسبوع

Technology Integration دمج التكنولوجيا	Use interactive torque simulations, projected force diagrams and digital calculation checks to investigate the effects of force, distance and angle.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: trigonometry, algebraic rearrangement, signed quantities and proportional reasoning. Engineering: levers and mechanical advantage.
External Assessment Integration دمج التقييمات الخارجية	AP Physics-style multi-step numerical reasoning, free-body diagrams, explanation of rotational equilibrium and experimental evidence.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	ATL: collaboration, critical thinking, communication and self-management. Emphasise safe handling of apparatus and respectful group roles.
Standards المعايير	Advanced Physics / AP Physics 1 rotational dynamics: torque magnitude and direction, net torque, and rotational equilibrium. NGSS science practices: Developing and Using Models; Using Mathematics and Computational Thinking.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	Why can a small force applied far from a pivot produce a larger turning effect than a larger force close to it?
Success Criteria معايير النجاح **(Measurable)**	**Lesson 1 — MUST:** Identify pivot, force and direction. **SHOULD:** Explain the turning effect of distance. **COULD:** Compare two lever arrangements with justification. **Lesson 2 — MUST**: Recall τ = rF sin θ and units N·m. **SHOULD**: Calculate torque for perpendicular forces. **COULD:** Solve and justify an angled-force problem. **Lesson 3 — MUST:** Distinguish clockwise and anticlockwise torques. **SHOULD:** Calculate net torque with signs. **COULD:** Determine an unknown force for equilibrium and justify. **Lesson 4 — MUST:** Record observations safely. **SHOULD**: Calculate torque from measured values. **COULD:** Evaluate evidence and recommend a design improvement.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Belonging / Global Understanding / ResponsibilityGlobal Understanding التفاهم العالميHeritage التراث
	المجال Area: UAE innovation, sustainable engineering and responsible design.
	التطبيق Implementation: Analyse hypothetical UAE bridge maintenance tools, cranes and door mechanisms; evaluate how safe lever design supports community infrastructure.